

ARTICLE

Which Workers Are Most Vulnerable to AI Disruption?

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This article uses a new AI measure to show how the risks and benefits of generative AI could be unevenly distributed across occupations, wages, education levels, demographic groups, and regions — including the Federal Reserve's Third District.

Artificial intelligence may serve as a transformative force for the U.S. economy, but its impact is likely to be uneven across subpopulations and geography. Will the benefits from AI be broadly shared or highly concentrated? Which workers are most vulnerable to generative AI disruption? These questions remain pressing concerns in public discourse, yet we still lack a clear understanding of how AI will alter economic opportunity, as **scholars and business leaders** debate how the technology will reshape the labor market and which jobs will be affected. Taking advantage of a new AI measure developed by Alex Martin Richmond at OpenAI (the **AI Jobs Transition Framework**, hereafter referred to as the Richmond

measure), we provide novel evidence of how the risks and benefits of AI-driven labor market disruption could be unevenly distributed across occupations, wage levels, demographic groups, and geography — suggesting that AI adoption may exacerbate, rather than alleviate, existing economic inequalities.

We integrate Richmond's job transition archetypes with detailed occupation, education, and demographic data[1] to provide a nuanced assessment of how AI-driven labor market disruption may affect different groups in the short term (two to three years). Based on the Richmond measure, we find that in the United States, 17 percent of jobs face higher short-term automation risk, while 11 percent are expected to grow owing to AI over this same time horizon. Jobs at higher risk of automation tend to be lower- to middle-wage positions with fewer educational requirements, although the lowest-wage jobs and those requiring no formal credential tend to be less affected. In contrast, occupations expected to grow are typically higher-wage roles requiring a bachelor's degree. These automation and growth risks fall unevenly across demographic groups: More than 20 percent of women's jobs are in high-automation-risk occupations, compared with just 10 percent of men's jobs, and Black and Hispanic workers have the smallest share of employment in AI-growth occupations, while Asian and White workers have the largest share. These patterns suggest that, based on the new Richmond measure, AI could deepen existing economic inequalities, with upside and downside risks distributed unevenly across demographic and socioeconomic groups.

Introducing Richmond's AI Jobs Transition Framework

To understand how generative AI will affect the labor market, researchers have created "exposure" measures (**Eloundou et al., 2023**) that assess the extent to which occupations could use AI to complete their tasks more quickly. Exposure scores, however, have important

limitations in studying AI's potential impact on the labor market. While they indicate how much workers might use (or benefit from) AI in their current roles, they cannot tell whether that capability will actually be deployed, nor can they shed light on whether tasks are likely to be displaced or augmented by AI.

Expanding upon exposure-only measures, the Richmond measure categorizes occupations[2] into four archetypes based on technical exposure, human necessity, and demand elasticity. Richmond's framework defines the three components as follows:

- Technical exposure is the share of an occupation's tasks that could be expedited by AI as measured in **Eloundou et al. (2023)**, providing an estimate of AI's capability to impact occupations.
- Human necessity determines the need for a human worker to perform the tasks associated with an occupation owing to regulatory, physical, or relational factors. Occupations with strong human necessity include, for example, accountants, registered nurses, and teachers.
- Demand elasticity measures how AI-related productivity gains may increase employment by lowering costs and increasing customer demand.

By incorporating human necessity and demand elasticity as additional measures, the Richmond measure captures AI's potential near-term labor market impacts rather than focusing solely on its technical capabilities. The latter two measures are quantified by prompting Chat-GPT 5.4 to measure the strength of the need for human workers and the sensitivity of demand to price changes for the associated occupational outputs.[3]

Table 1 outlines the four mutually exclusive archetypes defined by the Richmond measure. This analysis focuses on the first two: jobs at high automation risk and jobs that grow with AI. Jobs at high automation risk, such as customer service representatives or office clerks, are highly exposed to AI, require less human involvement and are unlikely to see increased demand as output prices fall. On the other hand, jobs that grow with AI, like software developers and graphic designers, also face high AI exposure; however, lower output prices could ultimately increase customer demand, creating more opportunities for these occupations. Occupations in the third category, jobs that will reorganize, are likely to experience a redistribution of tasks, with less obvious impact on growth over a two- to three-year time horizon.

It is important to underscore that the archetypes in this framework are not precise forecasts. The framework's structure is conceptually sound and more nuanced than raw "AI exposure" scores, but the estimates are still essentially informed guesses. Richmond cautions that the framework may "overstate substitution risk, understate augmentation and redesign, and miss the possibility that adoption unfolds in threshold-like or bundled ways rather than smoothly." Additionally, empirical evidence for widespread AI-driven job displacement remains scarce. By no means should the results be interpreted as predictions of AI-driven job loss or growth; instead, this exercise functions more as a scenario analysis grounded in a specific set of assumptions.

Table 1. Defining Archetypes from Richmond's Jobs Transition Framework

RICHMOND'S ARCHETYPE	DEFINITION	OCCUPATION EXAMPLES

Jobs at higher risk of automation	These occupations have high exposure, weak human necessity, and insufficient (or uncertain) demand elasticity to offset the direct labor-saving effect of AI-driven productivity gains.	Customer service representatives, data entry keyers, receptionists and information clerks, computer user support specialists
Jobs that grow with AI	These occupations have high exposure, but expanded output demand could more than offset the direct labor-saving effects of AI-driven productivity gains.	Graphic designers, software developers, health informatics specialists, electrical engineers
Jobs that will reorganize	These occupations have high exposure, strong human necessity, and insufficient demand elasticity to offset the direct labor-saving effect of AI-driven productivity gains.	Registered nurses, pharmacy technicians, human resources managers, supply chain managers
Jobs with less immediate change	These are occupations where the combination of the three measures do not point to a clear impact.	Retail salespersons, mental health counselors, heavy and tractor-trailer truck drivers, machinists

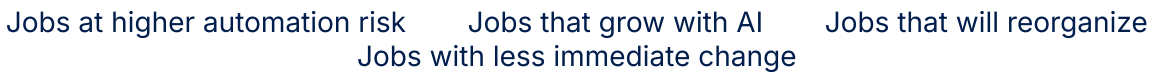
Source: Richmond (2026)'s **AI Jobs Transition Framework** (p. 14)

Lower- to Middle-Wage Roles Requiring Less Education Face Higher Automation Risk

Under the Richmond framework, jobs at a high risk of automation tend to be lower- to middle-wage roles requiring less education (high school or some college only). Over one-quarter of workers in occupations with median annual wages in the second and third quintiles (between $36,841 and $60,580), and 30 percent of workers in roles requiring just a high school degree face a high risk of automation (Figures 1 and 2). Some of the most common high-automation-risk roles include office and administrative support occupations like customer service representatives, office clerks, and secretaries and administrative assistants. These roles have high exposure to AI, low human necessity, and are unlikely to experience increased demand if the price of output decreases. Conversely, less than 1 percent of jobs in the lowest annual wage quintile and 2 percent of jobs without a formal education credential are at higher risk of automation. This is largely because most of these occupations require a human to deliver the service, such as food preparation workers, home health and personal care aides, or maids and housekeepers. This pattern is quite different from our previous research finding that higher-paying, higher-skilled occupations are most exposed to generative AI.

FIGURE 1

Share of U.S. Occupational Employment Across AI Archetypes by Median Wage Quintiles



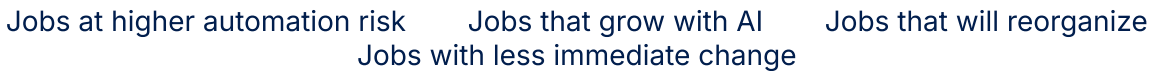
Source: Authors' calculations based on Richmond (2026) and BLS's 2025 Occupational Employment and Wage Statistics (OEWS)

On the other hand, jobs expected to grow with AI are most prevalent among high-wage occupations requiring a bachelor's degree. One-third of employment in occupations with a median annual wage of at least $82,221 are projected to grow in the short term because of AI, while 21 percent of employment in occupations with a median annual wage between $60,581 and $82,220 is projected to grow. Additionally, about 36 percent of employment in

occupations typically requiring a bachelor's degree is expected to grow with AI, at least in the near term. Some of the most common roles in this category include software developers, sales managers, and project management specialists. Although many of these roles may have high AI exposure and low human necessity, the demand for these occupations may increase as the price of output decreases, according to Richmond's measure.

FIGURE 2

Share of U.S. Occupational Employment Across AI Archetypes by Typical Education Level Needed



Source: Authors' calculations based on Richmond (2026) and BLS's 2025 Occupational Employment and Wage Statistics (OEWS) and 2025–2035 Employment Projections

Note: Shares may not add up to 100 because of rounding.

Women, Black, and Hispanic Workers Are Less Likely to Benefit from AI-Driven Growth

Women are more likely than men to hold jobs at a high risk of automation while less likely to hold jobs that grow with AI, based on the Richmond measure (Figure 3). Twenty-one percent

of women's employment is concentrated in high-automation risk occupations, more than twice the share for men. Additionally, women are less likely than men to benefit from AI-driven job growth. Only 9 percent of jobs held by women are projected to grow, compared with 13 percent of those held by men.

FIGURE 3

Share of U.S. Occupational Employment Across AI Archetypes by Gender

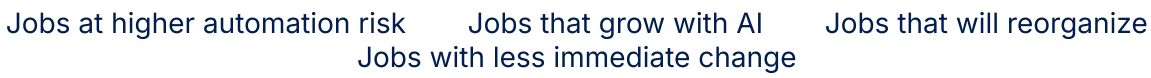


Source: Authors' calculations based on Richmond (2026) and 2024 American Community Survey five-year estimates via IPUMS

While automation risk is similar across race and ethnicity, Black and Hispanic workers are less likely to have jobs projected to grow with AI (Figure 4). Just 7 percent of jobs held by Black workers and 6 percent of jobs held by Hispanic workers are projected to increase, compared with 21 percent for Asian workers and 12 percent for White workers.

FIGURE 4

Share of U.S. Occupational Employment Across AI Archetypes by Race/Ethnicity



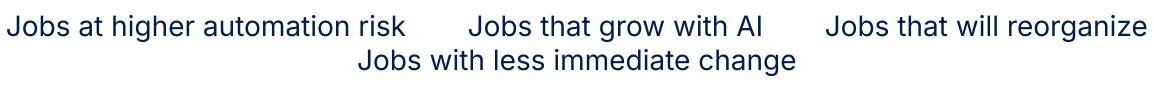
Source: Authors' calculations based on Richmond (2026) and 2024 American Community Survey five-year estimates via IPUMS

Third District Share of At-Risk Jobs Similar to National Share

About 17 percent of jobs in Third District MSAs face a high risk of automation, similar to the national average of 17.2 percent. The highest concentrations of jobs at high risk of automation are found in State College, PA; Trenton-Princeton, NJ; and Dover, DE. Vineland, NJ; Allentown-Bethlehem-Easton, PA-NJ; and Atlantic City, NJ have the smallest shares of employment at high risk of automation, but the gap in automation risk across markets is relatively small. Trenton-Princeton, NJ and Philadelphia-Camden-Wilmington stand out as having the largest shares of jobs projected to grow with AI. In contrast, remote and smaller MSAs tend to have fewer jobs projected to experience growth with AI, and there is a sizeable gap between regions with the highest and lowest shares of jobs with the potential for AI-driven growth (15.4 percent for Trenton-Princeton, NJ versus 3.7 percent for Vineland, NJ).

FIGURE 5

Share of Occupational Employment Across AI Archetypes within Third District MSAs



Source: Authors' calculations based on Richmond (2026) and BLS's 2025 Occupational Employment and Wage Statistics (OEWS)

Closing Thoughts

The potential impact of AI on inequality remains a pressing concern in public discourse, and this analysis suggests these concerns warrant careful attention. Our analysis based on a new AI typology suggests that automation risk could be concentrated among lower- to middle-wage occupations, while growth opportunities are concentrated in roles that require college degrees and pay higher wages. Women face twice the automation risk as men, and Black and Hispanic workers are underrepresented in occupations expected to grow because of AI. Additionally, smaller metro areas in the Third District are less likely to benefit from AI-driven job growth, potentially widening the economic gap between large metropolitan hubs and smaller regional economies.

Of course, important caveats apply when interpreting the results: Little empirical evidence currently supports the claim that AI is driving broad-based job displacement so far, and tangible large-scale inequality effects haven't been observed yet. Moreover, **there is disagreement among economists** over the degree and direction of AI's potential inequality effects. Notably, **recent research by David Autor** suggests that AI could possibly restore expertise to middle-skill workers, offering a more optimistic counterpoint to concerns about labor market polarization. As the technology matures, longer-term effects will require careful monitoring. Keeping all these caveats in mind, this exercise demonstrates that AI could accelerate existing disparities. Provided the assumptions behind the Richmond measure hold over time, lower-wage workers, women, and workers without a bachelor's degree may have lower access to AI-induced job growth. Without deliberate policy

interventions, AI may concentrate economic gains at the top while leaving vulnerable workers further behind.

Promising strategies may exist for mitigating the worst-case scenarios. **Recent research reveals an interesting divergence in preferred approaches**: Experts strongly favor targeted measures like worker retraining programs, while the general public supports both targeted interventions and broader solutions, including job guarantees and universal basic income. This gap suggests the need for inclusive policy dialogue that bridges expert analysis with public concerns. Ultimately, AI's impact on inequality is not predetermined. With proactive interventions, evidence-based policymaking, and continued monitoring of emerging effects, policymakers may be able to work toward an AI-enabled future that expands opportunity rather than concentrates advantage.

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[1] We merge Richmond's job transition archetypes with occupation data from the BLS's Occupational Employment and Wage Statistics (OEWS) and Education and Training Assignments at the six-digit SOC level. For the demographic analysis, we use data from the American Community Survey's Public Use Microdata Sample (PUMS) to determine the distribution of employment that falls under each archetype by race and gender.

[2] Because the Richmond measure uses O*Net occupations, which are more detailed than those in the BLS and ACS data, there are two instances (tour and travel guides; clinical technologists and technicians) where multiple O*Net occupations with different archetypes fall under one six-digit SOC occupation. In these two instances, we opt to classify these occupations as "jobs that will reorganize" versus "jobs with less immediate change."

[3] See Richmond (2026) for more information on the prompting methodology.

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